

Lab 4: Animations

Task 1

Your task is to make the `texture_anim.py` demo more general and robust. The code should be easily used with a different sprite sheet.

Specifically, only changing a few constants the demo code should work loading for example the `girl_running` image, displaying the animation for that sprite instead.

You are not allowed to change the window display dimensions, but you may need to use some scale and do some math.

Note that the girl texture sheet is 1024×128, which you do not need to hard-code but might be a good reference as you create your own example below.

Once your code works for both `scarfy` and the `girl_running`, search the web to find our own character. Using that texture is an additional and personal test that shows your code generalization.

Your character should have at least 4 cells, and be of good/decent resolution.

Make sure to remove background as you save an image from the web, the image needs to have transparency, i.e. an alpha channel.

Task 2

Using our `pong` lab starting code, use the `class Cat` as a template to augment it with a jumping feature. Please use something else than my bad resolution cat.

You will have to define a ground, from which a jump may start from and where it lands to stop.

We are not trying to implement a realistic jump, just sensible motion.

I used the following three sites to help my thinking

1. [Adding gravity](#)
2. [Implementing realistic gravity](#)
3. [Speed and gravity](#)

Those numbers at the bottom of the page are interesting. How do we understand them?

Submission

- Code for both task
- Video of the jump, with voice over explaining me how you implemented the feature; how you made sense of the number you used? and as they compared to what you saw on webpage for example